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# Requirements framework for IoT device authentication using behavioral fingerprinting

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ABSTRACT

The Internet of Things (IoT) has more and more been integrated into our work and life. However, besides the benefits brought by the recent advancements, there is an increasing challenge for securing IoT devices and networks. A common security mechanism is authentication. However, IoT devices are often resource-constrained which make the use of state-of-the-art encryption technologies infeasible. Therefore, researchers are trying to develop lightweight authentication methods. A promising example of this is the use of behavioral device fingerprinting. Still, the remaining problem with this technology is that it is unclear which feature sets are most feasible to implement device fingerprinting schemes in practical systems. In short, the current research body lacks clearly defined requirements. To overcome this issue, this research aims to design a requirements framework for IoT authentication schemes using behavioral device fingerprinting. To do so, Design Science Research is used, incorporating a systematic literature review. In the end, a requirements framework for behavioral device fingerprinting authentication is presented. The proposed framework features 20 requirements in the four categories High-level IoT, Fingerprint sophistication, Machine learning sophistication, and Attack resistance. We have demonstrated the application of the requirements framework in this article. It is believed that the proposed framework will help researchers and practitioners to develop better IoT authentication solutions.

## 1. Introduction

Smart devices have become a new way of life. The more technology is progressing, the more devices people own and use. But is this technological progression costing us more than it is benefiting us? Deloitte found in its 2023 *Connected Consumer Study* (Deloitte, 2023) that around one-third of its survey respondents had experienced at least one type of digital breach or scam in the previous year. The study also finds that the likelihood of a breach is higher for households with more smart devices. A smart home full of IoT devices can fall prey to "thousands of hacking attempts in a week".

### 1.1. Background

The term Internet of Things (IoT) describes a collection of (often small) devices connected on a network. Examples of IoT devices include sensors, RFID scanners, and factory machines. Data analysis in IoT is often performed on a cloud server; however, securing the data transmission to the server remains challenging. IoT devices are subject to multiple threats. As traditional networking communications, IoT communications are subject to eavesdropping, masquerading, and replay attacks (Rana et al., 2021). Additionally, IoT devices are specifically prone to physical attacks because they are often placed in publicly accessible spaces (Rao and Prema, 2020).

Proper authentication can solve many of these security threats (Mohammad Fneish et al., 2023). However, IoT devices are often resource-constrained with limited computational power, battery, and memory, making it impossible to rely on traditional authentication schemes (Esfahani et al., 2019). To overcome this issue, many different lightweight authentication schemes are proposed in the literature e.g., Rana et al. (2021), Esfahani et al. (2019), Gope and Sikdar (2019), Wazid et al. (2020), Zhou et al. (2019), Hammi et al. (2020), Shahidinejad et al. (2022) and Li et al. (2017).

The currently proposed solutions go in various directions. One of these directions is the implementation of Multifactor Authentication (MFA) e.g., Yang et al. (2024), E and Sheeja (2022), Sarbini et al. (2022) and Alsahlani and Popa (2021). Usually, multifactor authentication validates a user based on a combination of the following

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factors: something they have, something they know, and something they are. Modarres and Sarbishaei (2023) extends this definition with the following factors relevant to IoT: something the user does and somewhere the user is.

Research towards MFA for IoT devices proposes authentication through unique Device Fingerprint (DFP). DFP can be created from hardware and software features (Yadav et al., 2020), and also network features. One example of hardware features are Physically Unclonable Functions (PUF)s, which allow IoT devices to have a unique, hardware-originating fingerprint that cannot be spoofed by an attacker (Modarres and Sarbishaei, 2023). As an example of network features, Noura et al. (2019) proposes the usage of the characteristics of the different communication layers, such as the received signal strength at the physical layer, to create a DFP.

### 1.2. Design science research

This study follows Design Science Research (DSR) (Johannesson and Perjons, 2014). In accordance with this methodology, the practice investigated in this research is designing and developing behavioral fingerprinting schemes for resource-constrained IoT authentication with a focus on feature selection. More specifically, an extra layer of security should be added to high-risk IoT devices that communicate with an external server, such as smart doorbells and security cameras in a smart home, by constantly verifying the devices to recognize attackers or compromised devices.

### 1.3. Problem definition & research question

Designing feasible lightweight authentication schemes for IoT continues to be challenging, especially considering the balancing act between performance and the security requirements the solutions must perform.

One promising solution is device fingerprinting based on features from the existing network communication as one factor in the multifactor authentication schemes. However, selecting feasible features remains challenging (Xu et al., 2015). Moreover, another open question in this regard is the application of DFP in actual access control schemes (Bezawada et al., 2018). The underlying problem is the lack of a requirements framework that aids in feature selection, considers DFP in an IoT authentication scenario, and allows for comparison between different schemes proposed by researchers.

To mitigate this problem, this study's research objective is to *design a requirements framework for behavioral device fingerprinting schemes in the lightweight IoT authentication domain with a focus on feature selection.*

For this purpose, this research asks the following research question: *What are the requirements for successfully implementing device fingerprinting in lightweight authentication schemes for IoT devices?*

### 1.4. Research contribution

This research contributes to the current body of literature in the areas of lightweight IoT authentication and device fingerprinting by presenting a novel requirements framework for DFP using behavioral features of IoT devices. The proposed framework is helpful for both scholars, researching the potential IoT authentication schemes, and the industry, implementing DFP schemes in their products.

## 2. Related literature

This section presents the current literature on lightweight IoT authentication and device fingerprinting. Moreover, studies on the requirements for both are highlighted.

### 2.1. Lightweight IoT authentication

Due to IoT devices' inherent resource constraints i.e., small and less powerful processors, low amounts of memory, and operation on battery power, it is often infeasible to apply the state-of-the-art encryption algorithms in authentication schemes (Esfahani et al., 2019). Therefore, different scholars propose multiple lightweight authentication schemes appropriate for IoT devices.

Lightweight authentication schemes for IoT can be classified on various parameters, including the underlying technology employed. Esfahani et al. (2019), Wazid et al. (2020) and Zhou et al. (2019) all use hash functions and XOR operations. These techniques are less computationally intensive than asymmetric encryption algorithms, which are conventionally used for authentication and key exchange. Moreover, Rana et al. (2021) uses symmetric encryption with pre-shared keys while Shahidinejad et al. (2022) employ a fuzzy logic-based controller. What many of the currently proposed solutions have in common is that they rely on a trusted node in the network that manages or at least aids the authentication process (Esfahani et al., 2019; Wazid et al., 2020; Zhou et al., 2019; Shahidinejad et al., 2022). This increases the IoT network's complexity, thus limiting the application feasibility. For peer-to-peer key agreement, there have been lightweight proposals utilizing pre-synchronized time information (Wang, 2018; Wang et al., 2018). The American institute for standardization NIST ran a competition for lightweight cryptography for constrained environments in 2018 and announced the selection of the Ascon family in 2023 (NIST, 2024). Though Ascon will be the coming standard of NIST, it performs best only on energy consumption. A few other algorithms perform better on latency, throughput, memory occupation and gate area according to a comparison study made by van de Meent (2022). The selection of lightweight cryptography is thus largely dependent on performance requirements. In addition, how to integrate the newly proposed lightweight cryptography in an authentication scheme for distributed IoT is a challenge.

### 2.2. IoT authentication requirements

To guarantee adequate security, IoT authentication schemes must meet certain requirements.

In this regard, Tange et al. (2020) presents a comprehensive IoT security requirements framework. Although the framework covers IoT security as a whole, it also contains valuable requirements for IoT authentication schemes. Next to the requirements listed under the category *Authentication* (key distribution, mutual authentication, non-repudiation, anonymity & privacy, attestation), network security requirements such as latency & timeliness, and availability are crucial to be considered as well when designing authentication schemes. Moreover, the maintainability of the system must also be examined. Overall, Tange et al. (2020)'s framework has been quite general and has not studied the unique requirements for behavioral fingerprinting, such as fingerprint sophistication and machine learning.

More concrete requirements are found in technical specifications for IoT authentication (Internet Society, 2017; IoT Security Foundation, 2021), and web authentication (OWASP, 2015). Overall, these guidelines are either quite broad in their scope e.g., specifying access control patterns or system setups, or focus almost exclusively on password authentication. Nonetheless, some requirements stated are relevant for fingerprint-based authentication e.g., documentability (OWASP, 2015), unique authentication parameters (IoT Security Foundation, 2021), and preventing brute force (Internet Society, 2017). Existing requirements frameworks are discussed in more detail in Section 5.

### 2.3. Device fingerprinting

Device fingerprinting (DFP) describes the process of identifying a device by its unique software, hardware or network communication characteristics i.e., its fingerprint. This subsection explains different DFP architectures and applications. Moreover, the feasibility requirements discussed in the present literature are introduced.

#### 2.3.1. DFP schemes architectures

Current literature describes DFP schemes' main components in a similar fashion. Yadav et al. (2020) divide the DFP process into three steps: first, a fingerprint template is created; then, in production, a live fingerprint is derived from captured data (e.g. via network communication), which is then matched with the template in the third step, fingerprint verification. Bezawada et al. (2018) and Kumar and Paul (2023) follow a more linear approach, however arrive at the same conclusion. The fingerprint verification can be performed using a rule-based algorithm or a Machine Learning (ML) algorithm (Yadav et al., 2020). The latter can be arranged as supervised or unsupervised learning (Bezawada et al., 2018). Concerning the actual fingerprint creation, all three papers largely agree as they all mention the steps of feature extraction and fingerprint generation (Bezawada et al., 2018; Yadav et al., 2020; Kumar and Paul, 2023). Moreover, Bezawada et al. (2018) and Kumar and Paul (2023) prepend the action of information capture and source identification, respectively, while Yadav et al. (2020) highlight fingerprint storage and access as an essential factor.

#### 2.3.2. DFP applications

The currently proposed DFP solutions are designed for various application domains. While this paper exclusively focuses on IoT authentication, it is worth considering advancements from other application domains as it is common in the current literature body.

One application domain uses DFP for WIFI authentication (Xu et al., 2015; Hua et al., 2018). Hua et al. (2018), for example, use networking side-channel information to create a fingerprint used for WIFI access point authentication. On the other hand, Xu et al. (2015) present a DFP feasibility framework and research gaps in DFP for WIFI that are also relevant for IoT authentication applications of DFP.

Moreover, in their survey, Kumar and Paul (2023) present different applications for DFP in cyber–physical systems. Cyber-physical systems is different from IoT by emphasizing the interaction with physical processes instead of lightweight communication as in IoT. But there are many things in common between the two when it comes to the network infrastructure. Kumar and Paul (2023) has identified condition monitoring of device or machine health, tracing and forensic applications, and security as the main application fields of DFPs. Within security, they refer to device identification, PUFs, and threat identification. Similarly, Song et al. (2019) introduce a DFP scheme for IoT authentication that can detect device types with 92.3% accuracy. According to the scholars, similar solutions could not achieve the same. Hamad et al. (2019), Bezawada et al. (2018) propose similar solutions.

Ahmed et al. (2022) and Msadek et al. (2019) also present DFP solutions for IoT, however, from an attacker's perspective. Ahmed et al. (2022) for instance, exhibit a scheme that allows potential malicious actors to detect device vendors, device types, and even individual devices in the victims' network. While this poses a significant security threat to IoT rather than securing it, the scholars' methods are helpful in the domain of IoT authentication.

#### 2.3.3. Features

DFP can be based on many different feature types. Yadav et al. (2020) distinguish between the two broad categories of hardware and behavioral (software) features. For instance, Bezawada et al. (2018) look at data transmissions on the network layer since, according to the scholars, an attacker's device behaves differently compared to the legitimate IoT device. An example of this are the ports and patterns used by the device (Hamad et al., 2019). Hamad et al. (2019) add flow-based features as an additional distinguishing factor, providing IP destination and inter-arrival time as examples. Slightly differently, Chowdhury and Abas (2022) distinguish between a network packet, radio signal, and MAC layer features.

If we only look at network traffic, Hamad et al. (2019) has used Ethernet packet size, IP packet, header size, TCP payload size, TCP payload data offset, packet time to alive, packet inter arrival time, TCP window size in its DFP application for IoT authentication. Ahmed et al. (2022) uses a different list of features which include burst time, burst bytes, burst packets, burst delay, packet delay, external port list, unique IPs, packet size list, unique ports, request-reply pair list, protocols(%), total packets, hostname list, total bytes, IP list, unique packet length, packet in/out ratio, tcp flags, HTTP(S) traffic. For most features, Hamad et al. (2019) calculate minimum, maximum, first quartile, median, third quartile, mean, variance, and interquartile. Ahmed et al. (2022) follow a similar methodology in this regard.

#### 2.3.4. Feasibility

When using DFP in authentication schemes, feasibility criteria for both DFP solutions and features are essential to consider.

Bezawada et al. (2019) and Yadav et al. (2020), citing (Xu et al., 2015), present essential properties that every DFP solution must have. First, universality means guaranteeing broad, if not complete, feature availability. Moreover, uniqueness refers to the fact that the solution must be able to uniquely identify a device. Further, permanence expresses the requirement for features to remain stable i.e., not change much over time, while being feasible to collect (collectability). Finally, the fingerprint must be hard to forge to be used for authentication. In practice, accuracy seems to be the leading quality assessment criterion since this parameter is stated with most reviewed proposed solutions as their performance metric e.g., Ahmed et al. (2022) and Son et al. (2021).

In addition to DFP solution feasibility, Kumar and Paul (2023) propose an extensive feature quality assessment taxonomy. It can be observed that many of the metrics stated here are similar to those discussed above. This makes sense only considering that the aforementioned performance parameters also consider the requirements of individual features. For example, accessibility in Kumar and Paul (2023)'s framework has the same aim as universality and collectability. However, Kumar and Paul (2023)'s taxonomy seems very generic, except for a few direct references to DFP and CPS.

Considering the perspective of DFP being used by a malicious actor to scan IoT networks, Ahmed et al. (2022) investigate the generalizability of fingerprints. The scholars define generalizability as the extent to which the ML model produces similar results on different data. In this regard, they consider the temporal, spatial, and method impact. Although Ahmed et al. (2022) do not find a significant temporal impact, considering DFP from an authentication perspective this factor must be accounted for as a device's behavior can change over time e.g., through updates. Moreover, the spatial impact must be considered since IoT devices can be moved. In the end, any change in the device's behavior will result in false positives i.e., the actual device is thought of as a malicious device because the fingerprint does not match anymore.

### 2.4. Considerations framework

The results of this section are summarized in a considerations framework presented in Table 1. This framework lists factors to consider when designing DFP schemes for resource-constrained IoT devices, alongside a description and rationale for including the factors in the framework. The included factors are derived from literature and specifications from the areas of IoT authentication, web authentication, and DFP.

## 3. Methodology

This research utilizes Design Science Research (DSR) (Johannesson and Perjons, 2014) coupled with a systematic literature study to create a requirements framework for DFP authentication schemes for IoT devices. This section presents the methodological choices.

**Table 1**
Considerations Framework.

| Consideration | Source | Explanation | Rationale |
|---|---|---|---|
| Feature availability | Bezawada et al. (2018), Yadav et al. (2020), Xu et al. (2015) | To what extent are features available in the network traffic when required? | Features must be available when the fingerprint is created in production to guarantee accurate validation. |
| Feature stability | Bezawada et al. (2018), Yadav et al. (2020), Xu et al. (2015) | To what extent do features remain constant over time? | Features must remain constant compared to the time when the fingerprint was created. |
| Feature uniqueness | Bezawada et al. (2018), Yadav et al. (2020), Xu et al. (2015), IoT Security Foundation (2021) | To what extent is a feature unique? | Features must be unique for a device for identification purpose. |
| Fingerprint forgeability | Bezawada et al. (2018), Yadav et al. (2020), Xu et al. (2015) | To what extent is the fingerprint forgeable? | To protect data integrity and counter identity-theft attack, device authentication shall only use secure fingerprints that are difficult to forge. |
| Accuracy | Ahmed et al. (2022), Son et al. (2021) | How accurate can a device be identified? How accurate can false devices be identified? | If, in the end, the fingerprinting scheme decides which device gets authenticated and which does not, the scheme must be highly accurate. Otherwise, there is a chance of not authenticating legitimate devices (which reduces availability) or authenticating illegitimate devices (which compromises security). |
| Documentability | OWASP (2015) | To what extent is the fingerprinting process documented? | OWASP (2015) mentions documentability of authentication schemes as a key requirement. For example, during incidence response it helps to know who tried to get authenticated and how and why the system decided. |
| Timeliness | Tange et al. (2020), OWASP (2015) | How fast can a device be validated? | Malicious devices should be recognized quickly. |
| Device availability | Tange et al. (2020) | To what extent does the fingerprinting affect the devices availability? | Many false positives could lead to unnecessary down-times. |
| Device maintenance | Tange et al. (2020) | To what extent can the device be maintained (e.g., updated) without having to redo the fingerprint? | Related to feature stability. |
| Device reset | Internet Society (2017) | To what extent does the fingerprinting scheme allow for device resets? | The user should be able to reset device. |

### 3.1. Research strategy

This study follows a mixed research strategy approach. Firstly, DSR is used for developing and evaluating an artefact i.e., the requirements framework. In accordance with the description of the DSR cycle by Johannesson and Perjons (2014), this research performs the following DSR stages in an iterative manner: explicate problem, define requirements, develop artefact, and demonstrate artefact. The explicated problem has already been described in the first section (see Section 1.3). The remaining stages are elaborated upon in the results section (see Section 4.1, 4.3, and 4.4), respectively.

### 3.2. Data collection & analysis

In this study, DSR is used to design a requirements framework i.e., a model, for behavioral device fingerprinting in the lightweight IoT authentication context. For the development of this artefact, a systematic literature review is conducted to find relevant requirements for DFP authentication schemes and existing artefacts. A systematic literature review is an empirical study which uses academic publications as its core data input. Papers are carefully selected for examination based on strict criteria (see next section).

As the data analysis method, the articles are reviewed using a thematic analysis as described by Braun and Clarke (2006). This is an example of a qualitative data analysis. Braun and Clarke (2006) define thematic analysis as "a method for identifying, analyzing, and reporting patterns (themes) within data." (p. 79).

### 3.3. Sampling

According to Kitchenham and Brereton (2013), systematic literature review methodologies include inclusion, exclusion, and quality criteria based on which articles are filtered. Therefore, multiple criteria are defined. First, only articles from known academic literature providers such as IEEE, ACM, Springer, and Elsevier are included. Moreover, journal articles, as opposed to conference proceedings, are preferably selected. From the latter, only articles from the top 100 ranking journals are included. Further, the researcher ensured that the selected articles include the specific authentication requirements that are related to the considerations framework presented in Section 2.4.

## 4. Results

### 4.1. Artefact requirements

Per the DSR framework, this section presents the design requirements for the artefact developed in this study. Both functional and non-functional requirements are defined, where the former refers to requirements that describe the form and function of the artefact. At the same time, the latter delineates the supporting facets of the artefact such as ease of use. The artefact requirements are defined as follows:

I. *The artifact should outline specific requirements for behavioral device fingerprinting (functional).*
II. *The artifact should give researchers and practitioners clear guidelines on selecting feasible features for behavioral device fingerprinting (functional).*
III. *The artifact shall be structured comprehensibly so that it is easy to follow for people familiar with the domain (non-functional).*

In the following, we use thematic analysis to analyze data collected from a systematic literature review. The artefact requirements correlate with parameterization in thematic analysis by structuring key attributes necessary for IoT authentication. Among the thematic themes

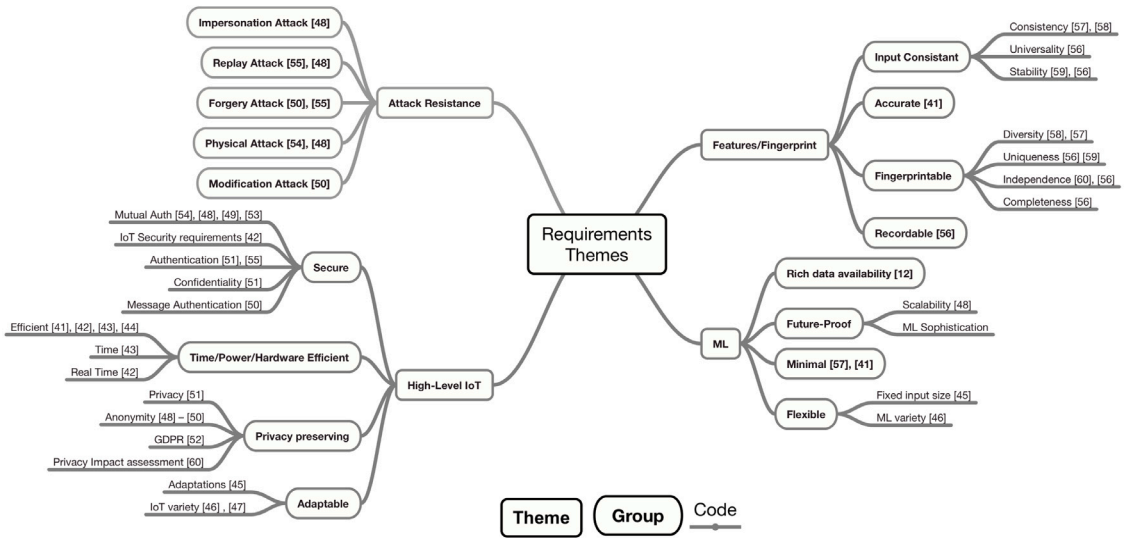


**Fig. 1.** Themes, groups and codes from the systematic literature review.

presented in Section 4.2.3, High-Level IoT Requirements (e.g. security, efficiency, privacy) align with broad IoT system expectations (Requirement I). Fingerprint Sophistication (e.g. accuracy, input consistency, recordability) corresponds to feature selection for device fingerprinting (Requirements I and II). ML Sophistication (e.g. rich data availability, minimal complexity, flexibility) ensures practical and scalable ML integration and ML is often the technique used to learn features (Requirement I). Attack Resistance (e.g. impersonation, replay, forgery attacks) maps to security vulnerabilities and mitigation strategies that raise requirements on on-device and inter-device authentication protocols (Requirement I). Overall, we provide a comprehensive structured framework that is easy to follow (Requirement III).

### 4.2. Systematic literature review

#### 4.2.1. Data collection

The literature search has resulted in 27 selected articles that meet the criteria defined in Section 3.3. 81% of the selected articles present novel solutions. The remaining articles are literature survey studies.

The selected articles are divided into 11 distinct areas. While most articles are directly related to DFP i.e., behavioral DFP or PUF, many articles are also associated with other IoT security categories such as smart vehicles, e-health, or IoT user authentication. Some articles are from different but related subject areas, such as biometrics.

#### 4.2.2. Data analysis

A thematic analysis is conducted to extract and analyze relevant data from the collected articles. In accordance with the guidelines given by Braun and Clarke (2006), the data is first screened and coded. More specifically, relevant passages from the text are excerpted and given a tag (code). This is an iterative process for finding new codes and aligning codes with each other. In this respect, code names are kept to a minimum number of words i.e., not exceeding four.

Subsequently, the coded excerpts are sorted into groups on a digital canvas. With this, similar tags are grouped together. Upon grouping, the group is given a name. For example, the codes *consistency, universality,* and *stability* are joint to form the group *input consistency*.

Hereafter, groups are once again connected into broader requirement themes. While the groups represent individual requirements, the themes serve as a broader identification of to which area and level of analysis the requirements belong. A visualization of all themes, including groups and codes can be found in Fig. 1.

#### 4.2.3. Findings

In the end, four themes emerge: High-Level IoT, ML Sophistication, Fingerprint Sophistication, and Attack Resistance. These represent specific requirement areas for behavioral DFP solutions. The groups that the themes comprise embody the specific requirements.

Considering these themes, it becomes apparent that they can be placed in a hierarchy in which High-Level IoT (as the name suggests) is placed on a broader level than the other three themes. Fingerprint Sophistication, ML Sophistication, and Attack Resistance are all concerned with one specific area of the IoT system that uses behavioral DFP as a continuous authentication method. Below, the emerged themes, as well as their underlying groups and connections to the literature, are explained in more detail.

*High-level IoT.* The systematic literature review revealed that IoT systems, in general, must be efficient, adaptable, privacy-preserving, and secure.

In that regard, the need for efficiency is comprehensively described in the literature. It includes the requirement for power consumption to be as low as possible (Bhuva and Kumar, 2023), for authentication to happen in real-time (Wang et al., 2023b; Batool et al., 2022), and the system to be performant (Baghel and Prakash, 2022).

Moreover, an IoT software must be adaptable since there exists a vast amount of different kinds of IoT devices (Jmila et al., 2022; Sanchez et al., 2021; Feng et al., 2023).

Additionally, IoT devices should be privacy preserving. The user should not be traceable (Iqbal et al., 2020; Ding et al., 2021; Zhong et al., 2021), and their data should not be accessible to anyone else (Zhu et al., 2018; Iqbal et al., 2020). Also, concerning privacy, there is an obligation to comply with legal requirements such as GDPR (Alhirabi et al., 2021).

Finally, IoT devices should be secure. Often-cited requirements include mutual authentication (Ding et al., 2021; Iqbal et al., 2020; Zerrouki et al., 2023; Shang et al., 2024) and authentication in general (Zhu et al., 2018; Cui et al., 2023; Zhong et al., 2021).

*Fingerprint sophistication.* Although other requirements must also be considered, it can be argued that requirements around the actual device fingerprint, its features, and the underlying network transmission information are the core areas of behavioral DFP requirements. Regarding this area, the thematic analysis finds that input consistency, fingerprintability, recordability, and accuracy are essential requirements.

In this regard, being input consistent means that the characteristics of the fingerprint must stay the same over time (Dong et al., 2024; Yu

et al., 2020; Chen et al., 2022). Moreover, the source information must be "universally present in the transmission signals" (Dong et al., 2024, p. 5).

Furthermore, to achieve a valid and comprehensive fingerprint, the features and their underlying information must be unique within a device (Dong et al., 2024; Wang et al., 2021) and diverse between devices (Chen et al., 2022; Yu et al., 2020). Moreover, features must be independent of the content that is transmitted (Dong et al., 2024; Liang et al., 2020).

In addition to this, the data for the feature must be reliably recordable (Dong et al., 2024).

Finally, the resulting fingerprint solution must be highly accurate to differentiate between authorized and non-authorized devices.

*ML sophistication.* Another crucial requirement area concerns the fingerprinting algorithms that often utilize machine learning. Here, the systematic literature review reveals that high data availability, minimalism, flexibility, and future-proofness are essential characteristics of successful behavioral DFP solutions.

ML and especially deep learning (DL) algorithms require large amounts of data to work accurately (Yang et al., 2024). Hence, for fingerprinting algorithms to succeed, they require large amounts of training and testing data. This includes data on different features as well as on other devices.

Moreover, keeping in mind the resource constraints of the IoT devices in focus, ML algorithms should be adequately minimal. This includes low algorithm complexity (Bhuva and Kumar, 2023) and using as few network protocols as possible (Yu et al., 2020).

Further, ML algorithms must be flexible in the sense that they can deal with varying data. Hereby, developers must overcome the challenge related to the requirement of fixed input sizes by ML algorithms (Jmila et al., 2022) and different input sizes by different ML methods (Sánchez et al., 2022).

Finally, ML algorithms should be future-proof. They should be scalable (Iqbal et al., 2020) and must be sophisticated enough to handle the ever-growing complexity of data.

*Attack resistance.* Attack resistance can be viewed from two different perspectives. First, one can look at it considering attacks. Alternatively, one can consider resistance from the standpoint of security principles.

Regarding the former perspective, many (types of) attacks are listed in the literature. These include impersonation attacks (user, gateway, IoT device) (Chaudhry et al., 2021), replay attacks (Chaudhry et al., 2021; Iqbal et al., 2020; Cui et al., 2023), forgery attacks (Cui et al., 2023; Zhong et al., 2021), physical attacks (Shang et al., 2024; Chaudhry et al., 2021), and modification attacks (Cui et al., 2023; Zhong et al., 2021).

Concerning the perspective of security principles, the current literature body provides only few required properties that go beyond the commonly quoted principles of confidentiality, integrity, and availability. The first property discovered during the systematic literature review is unlinkability, which describes the state in which an attacker cannot recognize a message's sender (Ding et al., 2021; Zhong et al., 2021). Another stated requirement is freshness that adds a timestamp to the message to circumvent replay attacks (Zerrouki et al., 2023). Finally, backward security is an additional principle quoted in which an attacker cannot decrypt messages with an encryption key obtained later (Ding et al., 2021; Zhong et al., 2021).

### 4.3. Artefact: Requirements framework

Table 2 presents the requirements framework for behavioral DFP authentication schemes, which is the final deliverable of this research study. In DSR terminology, this artefact can be described as a model, since it outlines the characteristics of a perfectly designed behavioral DFP authentication scheme. The framework exhibits a baseline for behavioral DFP authentication scheme research and development along with providing a common ground for comparison between competing schemes.

The artefact comprises 20 unique requirements, divided into four broad categories. Hereby, it must be noted that the High-Level IoT category includes broad requirements that all kinds of IoT device should follow. Meanwhile, the remaining three categories focus on specific areas of an IoT node using behavioral DFP. In this context, fingerprint and ML sophistication are directly linked to DFP, while attack resistance focuses on the kinds of attacks that the authentication should prevent.

The artefact is developed primarily based on the information gathered during the systematic literature review. The themes that derive from this analysis form the basis for the requirement framework's categories. The theme's underlying groups are the basis for the individual requirements. These are supplemented by the considerations introduced in Section 2.4.

### 4.4. Artefact demonstration

The artefact is demonstrated with the help of two behavioral DFP authentication solutions for IoT devices, presented by Hamad et al. (2019) and Bezawada et al. (2018) respectively. These two studies are chosen because they both present novel IoT DFP solutions. Moreover, the two papers focus on DFP for IoT authentication purposes and behavioral DFP specifically, which aligns with the study area of this study and the presented framework. As part of the analysis, the papers in question are scanned for information that helps determine whether or not the requirements are met. The results are presented in Table 3.

It becomes apparent during the analysis that the information provided in the articles is insufficient to rate the solutions on all the framework's requirements. Therefore, in some cases, it is impossible to determine whether a requirement is met or not met or if the necessary data is just not made available.

During the analysis, it becomes evident that both solutions primarily focus on features — accuracy and ML. Although it seems to be a common practice to discuss attack resistance of the presented solution in papers about hardware DFP and other lightweight authentication schemes, this is not done in the highlighted articles. Similarly, performance figures regarding the computational costs and the related costs of time and energy are not mentioned, despite this being standard practice in the field. Probably, the reason for this is the fact that here, the attention is on the fingerprint itself rather than a complete authentication scheme.

We consider the high-level IoT requirements are partially met by both solutions. Both papers discuss fingerprinting security to prevent device impersonation. Both methods use passive fingerprinting techniques but do not explicitly address privacy protection. Both approaches work on diverse IoT devices with different network protocols and thus are adaptable. Both papers provide detailed descriptions of their methods, datasets, and results, and are thus well documented.

Regarding fingerprint sophistication, both solutions meet most requirements. However, both techniques rely on network behavior consistency for device fingerprinting but some features like packet payload, and TCP window size may easily change overtime. Hence, a specific device might not be clearly identified among its similar peers. As a result, diversity i.e., inter-device uniqueness, is not necessarily provided. This perhaps explains why the solutions only manage to distinguish device types, not specific devices.

Regarding ML sophistication, both solutions collect large datasets and rich data are available for network traffic based features. There is not much information about future-proof but new IoT protocols may require retraining of models.

**Table 2**
Artefact.

| Requirement | Definition | Source |
|---|---|---|
| *High-Level IoT* | | |
| Secure | IoT devices employing continuous authentication through behavioral DFP should be secure. This includes using mutual authentication, preserving confidentiality and integrity and having adequate attack resistance. | Ding et al. (2021), Iqbal et al. (2020), Zerrouki et al. (2023), Shang et al. (2024), Zhu et al. (2018), Cui et al. (2023), Zhong et al. (2021) |
| Efficient | IoT devices employing continuous authentication through behavioral DFP should work efficiently concerning execution time, power consumption and hardware efficiency. | Bhuva and Kumar (2023), Wang et al. (2023b), Batool et al. (2022), Baghel and Prakash (2022) |
| Privacy preserving | IoT devices employing continuous authentication through behavioral DFP should preserve the users privacy. | Iqbal et al. (2020), Ding et al. (2021), Zhong et al. (2021), Zhu et al. (2018), Alhirabi et al. (2021) |
| Adaptable | Behavioral DFP solutions should be adaptable so they work with a wide range of different IoT devices. | Jmila et al. (2022), Sanchez et al. (2021), Feng et al. (2023) |
| Documented | Behavioral DFP authentication processes should be well documented. | OWASP (2015) |
| Available | IoT devices employing continuous authentication through behavioral DFP should remain available for operations most of the time. | Tange et al. (2020) |
| *Fingerprint Sophistication* | | |
| Input Consistent | The input information for features of behavioral DFP solutions should remain consistent over time. | Dong et al. (2024), Yu et al. (2020), Chen et al. (2022), Bezawada et al. (2018), Yadav et al. (2020), Xu et al. (2015) |
| Accurate | The DFP solution should be accurate in the sense that it does not show too many false positives and false negatives. | Ahmed et al. (2022), Son et al. (2021), Bhuva and Kumar (2023) |
| Fingerprintable | The circumstances in the network communication allow for the creation of a viable fingerprint. This includes that the source information is diverse, universal, and unique. | Dong et al. (2024), Wang et al. (2021), Chen et al. (2022), Yu et al. (2020), Dong et al. (2024), Liang et al. (2020) |
| Recordable | The source information for the features is able to be recorded by means that are either already available or available through adequate extra effort. | Dong et al. (2024) |
| Unforgable | The resulting fingerprint is computationally infeasible to forge. | Bezawada et al. (2018), Yadav et al. (2020), Xu et al. (2015) |
| *ML Sophistication* | | |
| Rich Data Availability | A large amount of data should be available to train the fingerprinting ML algorithm. | Yang et al. (2024) |
| Future-Proof | ML fingerprinting algorithms should be future-proof, meaning that they are scalable for future data demands and sophisticated enough to create viable fingerprints from complex feature sets. | Iqbal et al. (2020) |
| Minimal | ML fingerprint algorithms should follow the minimization principle. They should calculate and validate fingerprints which are computationally inexpensive as feasible. | Bhuva and Kumar (2023), Yu et al. (2020) |
| Flexible | ML fingerprint algorithms should be flexible in a sense that they can deal with future feature expansion and varying ML-methods. | Jmila et al. (2022), Sanchez et al. (2021) |
| *Attack Resistance* | | |
| Impersonation attack | An attacker is not able to spoof a user, IoT device, server, or gateway by impersonating as another node. | Chaudhry et al. (2021) |
| Replay attack | An attacker is not able to resend messages from earlier authentication communication to gain authentication status. | Chaudhry et al. (2021), Cui et al. (2023) |
| Forgery attack | An attacker is not able to get authenticated by forging an authentication message including the fingerprint. | Cui et al. (2023), Zhong et al. (2021) |
| Physical attack | An attacker is not able to get authenticated with the server by having access to the IoT device. | Shang et al. (2024), Chaudhry et al. (2021) |
| Modification attack | An attacker is not able to change an authentication message it their advantage. | Zhong et al. (2021) |

There are no explicit discussion on attack resistance in both solutions.

## 5. Comparison with existing solutions

Although no comprehensive requirements framework for device fingerprinting in lightweight IoT authentication exists, some frameworks are available that cover specific requirement areas. Therefore, this section presents existing requirements framework related to IoT, DFP, security, and authentication. Table 4 illustrates the comparison of these solutions based on the inclusion of themes identified during the systematic literature review (see 4.2.3 for more details).

### 5.1. Literature

There are few literature that provide requirement frameworks on DFP-based IoT authentication. One example of this is Bezawada et al. (2019), who highlight *universality, uniqueness, permanence*, and *collectability* as essential requirements for device fingerprinting schemes. According to this framework, every feature should be available in all target devices, every device's fingerprint should be unique, features should remain stable, and the features' source information should be accessible. While these requirements cover the most essential aspects of the fingerprinting itself, this framework neglects broader IoT requirements as well as ML algorithm and security requirements.

**Table 3**
Artefact implementation at the example of Bezawada et al. (2018) and Hamad et al. (2019).

| Requirement | Hamad et al. (2019) | Bezawada et al. (2018) |
|---|---|---|
| *High-Level IoT* | | |
| Secure | (✓) | (✓) |
| Efficient | – | – |
| Privacy preserving | (✓) | (✓) |
| Adaptable | ✓ | ✓ |
| Documented | ✓ | ✓ |
| Available | – | – |
| *Fingerprint Sophistication* | | |
| Input Consistent | (✓) | (✓) |
| Accurate | (✓) | (✓) |
| Fingerprintable | (✓) | (✓) |
| Recordable | ✓ | ✓ |
| Unforgable | – | – |
| *ML Sophistication* | | |
| Rich Data Availability | ✓ | ✓ |
| Future-Proof | – | – |
| Minimal | – | – |
| Flexible | – | – |
| *Attack Resistance* | | |
| Impersonation attack | – | – |
| Replay attack | – | – |
| Forgery attack | – | – |
| Physical attack | – | – |
| Modification attack | – | – |

× := not met; (✓) := partially met; ✓:= met; – := no information available/not applicable.

**Table 4**
Existing solutions mapped against requirement themes.

| Solution | IoT | ML | DFP | Resistance |
|---|---|---|---|---|
| Bezawada et al. (2019) | × | × | ✓ | × |
| Tange et al. (2020) | ✓ | × | × | (✓) |
| ISOC | ✓ | × | × | (✓) |
| IoTSF | ✓ | × | × | (✓) |
| This paper | ✓ | ✓ | ✓ | ✓ |

× := not present; (✓) := partially present; ✓:= present.

Moreover, Tange et al. (2020) present a broad IoT requirements taxonomy, which includes requirements from the areas of network security, authentication, access control, maintainability, and resilience. Although this framework is expansive, it does not go into much detail on each of the areas. Additionally, it focuses on IoT as a whole, and therefore, it does not explicitly include requirements on DFP.

In summary, Bezawada et al. (2019) and Tange et al. (2020) discuss general IoT security concerns and touch on authentication without offering a concrete structured framework, while we explicitly structure 20 distinct requirements under four themes and make it more applicable for adoption.

### 5.2. Standards

Different organizations e.g., ISOC and IoTSF, publish technical specifications (standards) on IoT authentication requirements. For example, the *OTA IOT trust framework* (Internet Society, 2017) and the *IOT security assurance framework* (IoT Security Foundation, 2021) both include sections on authentication. While these standards have some relevance for DFP schemes through broad IoT requirements, they are often focused on password requirements and access control setups.

More specifically, the *OTA IOT trust framework* provides a requirement structure that is divided into security, user access, privacy, and notifications, and covers user credentials, password policies, encryption, but not device fingerprinting. The *IOT security assurance framework* uses an assurance questionnaire & classification system for security. It covers authentication & authorization but lacks a dedicated device fingerprinting approach. Both frameworks have not considered the requirements for ML integration. In contrast, we provide the most comprehensive, structured, and fingerprinting-specific requirements framework for IoT authentication, integrating considerations on high-level requirements, features selection, ML integration and attack resistance.

### 5.3. Assessment tools

Many scholars use security verification tools for authentication schemes such as ProVerif (Iqbal et al., 2020; Ding et al., 2021; Wang et al., 2023a) or VerifPal (Zerrouki et al., 2023). Both ProVerif and VerifPal provide formal verification of cryptographic protocols. While these tools offer built-in checks for common security attributes like confidentiality and integrity, they do not state specific requirements. They do not consider feature requirements for DFP, do not consider ML, and are not directly comparable to our presented requirements framework. However, the attack resistance requirements can be potentially verified by using these tools if an authentication protocol that follows the requirements framework in this article is proposed and evaluated in the future.

## 6. Discussion

This study presents a requirements framework for behavioral device fingerprinting solutions that are applied in the context of resource-constrained IoT devices. This section discusses this artifact (i.e. the proposed requirements framework) in light of the current literature and existing solutions. Moreover, limitations are outlined. Finally, a conclusion is given in which the research question of this study is answered.

### 6.1. Requirements framework

This research introduces a novel requirements framework for DFP solutions related to lightweight IoT devices. It comprises 20 unique requirements, which are grouped into four categories. In this section, some aspects of the artefact are discussed.

#### 6.1.1. Privacy concerns

The presented framework states privacy preservation as a high-level IoT requirement. This includes the information that forms the feature set of the fingerprint. Here, a dilemma is created between privacy concerns and the potentially improved security through using more device-specific data for the fingerprint. Of course, mitigation strategies to preserve privacy can be employed. Nonetheless, this ethical aspect must be considered during feature selection.

#### 6.1.2. Novelty

The unveiled artefact is a novel solution to the problem of the lack of a requirements framework that aids the design of behavioral DFP solutions. The novelty expresses itself through two things. First, the presented artefact includes new requirements compared to the existing solutions. For instance, attack resistance against specific attacks is not part of any existing solution presented in Section 5 (also see Table 4). Likewise, ML sophistication is not part of any current solution either. This study introduces this requirement category along with crucial requirements in its framework. Moreover, the introduced artefact includes requirements that are future-oriented. Existing solutions are merely concerned with getting DFP/IoT authentication to work once. In addition, this solution considers scalability and adaptability issues as well.

The second reason why the introduced framework represents a novel solution is because it is a holistic framework. Using the results of the systematic literature review, the considerations framework, and brainstorming, different frameworks are combined. The resulting framework comprises requirements from various perspectives. This allows researchers and practitioners to design DFP solutions that work in the greater context of IoT systems.

### 6.2. Future research

Although this paper already introduces a complete framework, more research is needed to validate and possibly extend or modify the presented artefact. Researchers developing a behavioral DFP solution for authentication could, for instance, use this framework and, in a parallel study, assess its usability. Additionally, once the framework has been published for some time, scholars could conduct user satisfaction and adoption studies to measure the success of the delivered artefact.

### 6.3. Conclusion

This study asks about the requirements for successfully implementing device fingerprinting based on hardware and network communication features in lightweight authentication schemes for IoT devices. Based on the conducted research, it can be concluded that researchers and practitioners should view the problem from multiple perspectives when designing behavioral DFP authentication solutions for resource-constrained IoT devices. More specifically, they should consider requirements in the areas of DFP, attack resistance, and ML while not losing focus on fundamental IoT requirements. For this purpose a requirements framework is developed using the Design Science Research strategy that includes requirements from all the four areas. With this deliverable, this study is contributing to the ongoing research in the field of IoT authentication. Scholars can make use of this framework to develop better behavioral DFP authentication schemes. This paper even benefits society as a whole indirectly since the contributions that will be based on this study will ultimately make IoT devices, which already are a constant factor in our lives more secure.

## CRediT authorship contribution statement

**Ole Höfener:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Qinghua Wang:** Writing – review & editing, Supervision, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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